Rayane Dakhlaoui

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Professional Experience

AI Researcher Intern @ Siemens | ***Munich, Germany*** *Apr. 2026 – Sept. 2026*

- Research internship within Siemens Foundational Technologies, developing geometric deep learning models for 3D understanding of industrial CAD parts.
- Built a causal data-generation pipeline to curate and annotate large-scale synthetic 3D training data with semantically valid labels.
- Designed self-supervised pre-training via a Joint-Embedding Predictive Architecture (JEPA), with latent-space prediction over a point-cloud backbone (Utonia) and few-shot transfer to scarce real data.
- Combined heterogeneous graph neural networks (Graph Transformers) with point-cloud representations for structured 3D feature recognition.

Academic Education

PSL Research University *Sept. 2025 – Dec. 2026*

MScR IASD (Artificial Intelligence, Systems, Data) - Mathematics Track

- Top-tier French research-oriented program in Applied Mathematics, Learning & Modeling
- **Relevant coursework:** Reinforcement Learning · Advanced Machine Learning · Dimension reduction and manifold learning · High-Dimensional Statistics · Bayesian Statistics · Deep Learning for Computer Vision · Optimal Transport for ML · Machine Learning with Kernel Methods · Large Language Models for Code & Proof · 3D Point Cloud and Modeling · Convex Optimization

Télécom Paris - Institut Polytechnique de Paris *Sept. 2023 – Dec. 2026*

Engineering Degree (M.S.), Applied Mathematics and Computer Science

- Top-tier French Engineering school, Leader in the field of digital technology
- **Random Modeling Track:** Martingales · Markov chains · Stochastic Processes (Brownian motion, Poisson process) · High-Dimensional Probability · Numerical analysis (Newton, Monte-Carlo)
- **Signal Processing for AI Track:** Statistics · Optimization for ML · Time-Series Analysis · Machine Learning · Deep Learning (CNN, RNN, VAE, GAN, Transformers) · LLM · Audio & Speech processing · Databases

Preparatory Class for French Grandes Écoles *Sept. 2021 – June 2023*

Scientific Track PSI (eq. BSc. in Mathematics and Physics) - Lycée Condorcet, Paris IX*

- Highly selective 2-year intensive program in Mathematics & Physics; ranked top 8% nationally

Notable Projects

GAN Latent Space Tuning | **Github**

- Studied the precision–recall trade-off of f-GANs using JS, KL, RKL and BCE objectives on MNIST
- Implemented Soft Truncation and Discriminator Rejection Sampling to improve post-hoc sample quality

Hackathon IBM x Qubit Pharmaceuticals x Margo | **Github**

- Finalist (Top 4/25 teams) in a 56-hour hackathon on in-silico prediction of hERG inhibition from molecular structure
- Developed ensemble models combining GNN, XGBoost, and ChemBERTa for toxicity prediction

Research Project on Collaborative Filtering | **Github**

- Developed a framework implementing Multiple Kernel Matrix Factorization for collaborative recommender systems
- Built on top of scikit-learn and accelerated computations using Numba

Research Project on 2PAC Blockchain Consensus | **Research Paper** | **Github**

- Implemented and evaluated the 2PAC blockchain consensus in collaboration with Pr. Matthieu Rambaud
- Simulated the consensus under extreme conditions, managing node corruption and latency using sockets

Skills

Programming: Python · SQL · TypeScript · C · LaTeX
Tools & Frameworks: Numpy · Pandas · Scikit-learn · TensorFlow · Keras · PyTorch · Git
Languages: French (Native) · English (Full professional proficiency) · Arabic (Intermediate) · German (Beginner)